ENTITY RELATIONSHIP DIAGRAM

----Where data meets the structure

<u>ER DIAGRAM (entity relationship):</u>

An ER Diagram (Entity–Relationship Diagram)in Database Management System is a graphical representation of entities, attributes, and relationships in a database. It is used to design and visualize the structure of a database before creating the tables.

Helps in:

Structuring data

Avoiding redundancy

Better understanding of system

<u>Components of an ER Diagram</u>

- Entity
- Attribute
- Relationship
- Cardinality

ENTITY

An **entity** is a real-world object or concept such as a person, place, company, or concept that can be uniquely identified and stored in a database.

Each entity is described by a set of attributes that represent its properties

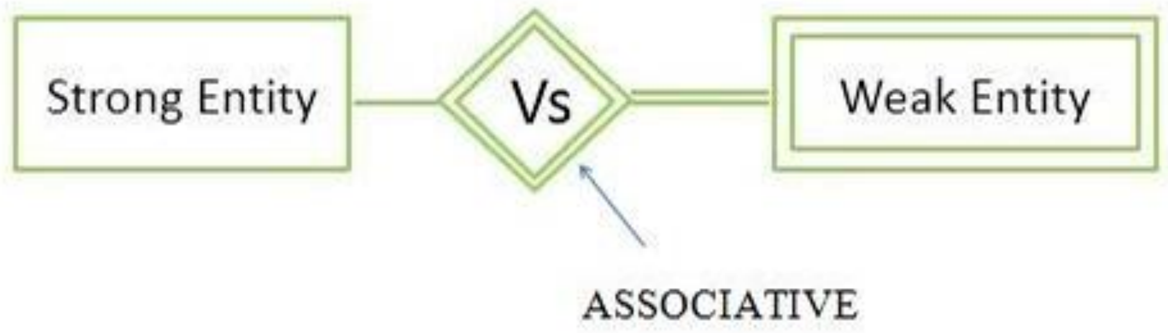
Entities are represented as tables in a database, where each row corresponds to an instance of the entity and each column represents an attribute

Types:

Strong Entity – Has its own key

Weak Entity – Depends on another entity

Associative Entity -represents the relationship between two or more entities



ATTRIBUTES

Attributes are the properties or characteristics of an entity. They describe the details of the entity and provide specific information about it. Each attribute corresponds to a column in a database table. Attributes have values for each entity instance (record).
EX: Name, Age, ID, and Address.

TYPES OF ATTRIBUTES:

- Simple Attribute – Cannot be divided further (e.g., Age, ID).
- Composite Attribute – Can be divided into smaller sub-parts (e.g., Full Name → FirstName + LastName).
- Derived Attribute – Value can be derived from other attributes (e.g., Age from Date Of Birth).
- Single-Valued Attribute – Holds only one value for an entity (e.g., Gender).
- Multi-Valued Attribute – Can hold multiple values for an entity (e.g., Phone Numbers).
- Key Attribute – Uniquely identifies an entity (e.g., StudentID).

RELATIONSHIPS:

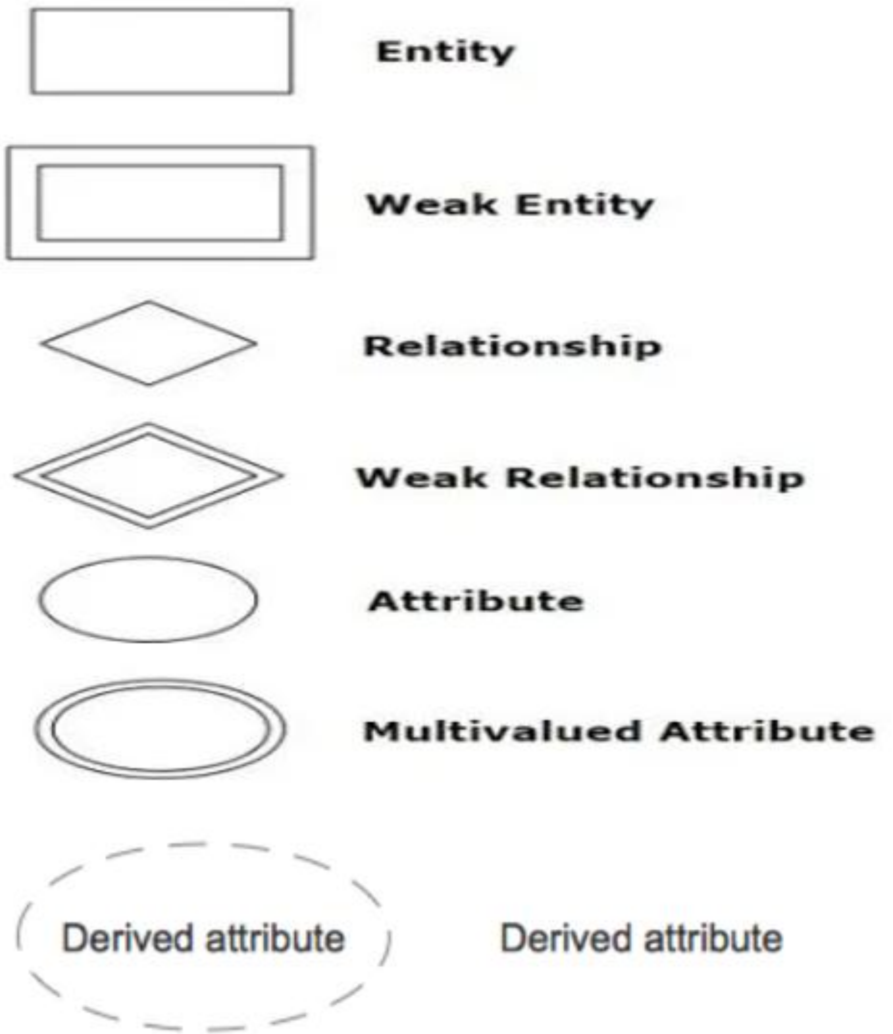
A relationship refers to the association or connection between two or more entities (tables) in a database. Relationships show how data in one table is linked to data in another table using keys such as primary keys and foreign keys.

A relationship in DBMS is the logical connection between tables that helps organize and retrieve related data efficiently.

TYPES OF RELATIONSHIPS:

- One-to-One Relationship (1:1): A relationship where one record in a table is related to only one record in another table. Example: One person has one passport.
- One-to-Many Relationship (1 : N): A relationship where one record in a table can be related to multiple records in another table. Example: One department has many employees.
- Many-to-One Relationship (N :1) : A relationship where many records in one table are related to one record in another table. Example: Many students belong to one college.
- Many-to-Many Relationship (M: N) : A relationship where many records in one table are related to many records in another table. Example: Students can enroll in many courses and each course can have many students.

SYMBOLS USED IN ER DIAGRAM



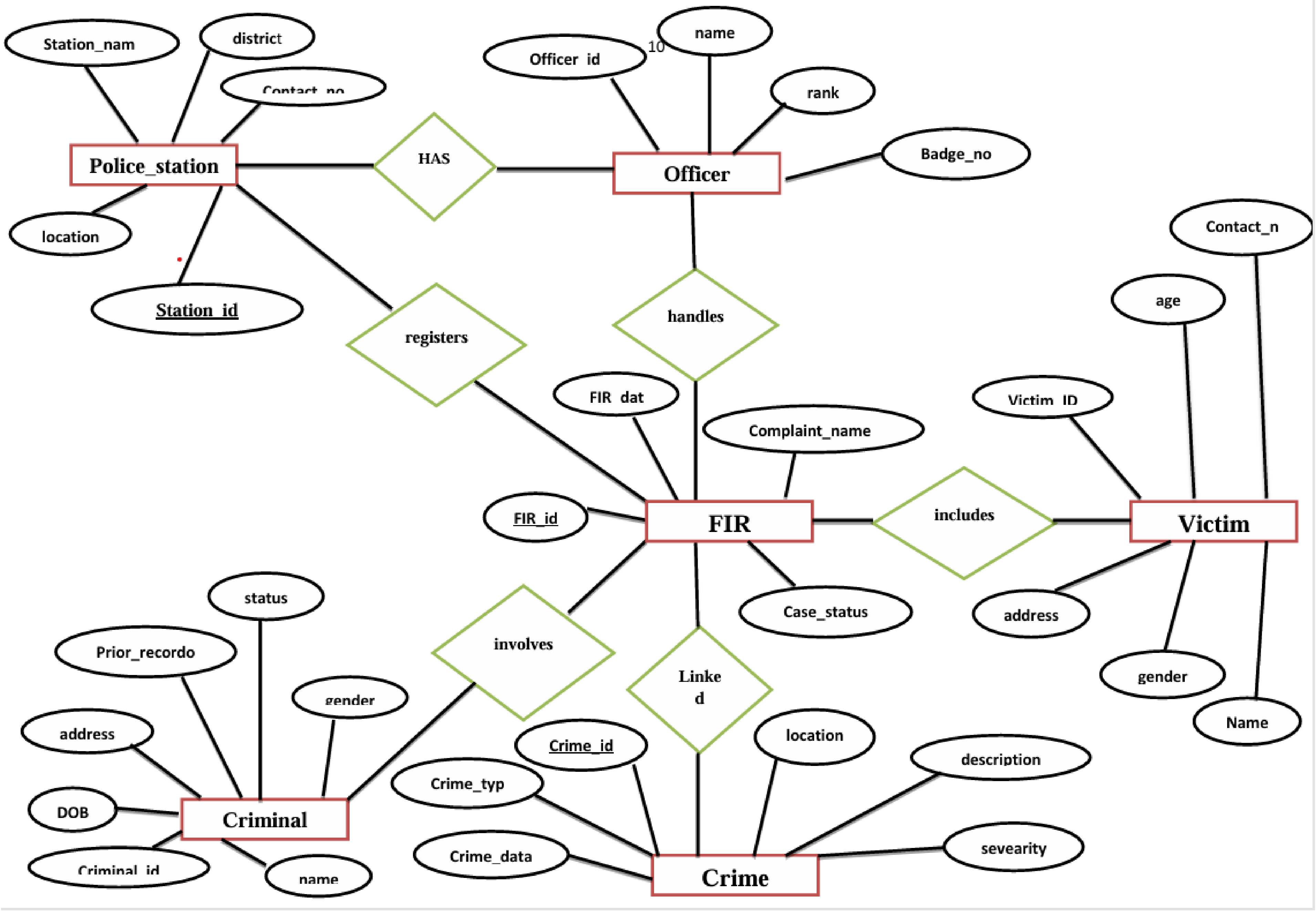
Let us consider the example ***CRIME RECORD MANAGEMENT SYSTEM***

ENTITIES SUMMARY:

- POLICE_STATION (station_id PK, station_name, location, district, contact_no)
- OFFICER (officer_id PK, name, rank, badge_no, station_id FK)
- FIR (fir_id PK, fir_date, complainant_name, station_id FK, crime_id FK, officer_id FK, case_status)
- CRIME (crime_id PK, crime_type, crime_date, location, description, severity)
- CRIMINAL (criminal_id PK, name, dob, gender, address, prior_records, status)
- VICTIM (victim_id PK, name, age, gender, contact_no, address)

From the relationships you gave, the verbs (relationship words) are:

- POLICE_STATION– OFFICER→ has
- OFFICER –FIR→ handles
- CRIME –FIR→ linked to
- FIR – VICTIM →includes
- FIR –CRIMINAL→ involves
- POLICE_STATION – FIR → registers


Station_nam
district
Contact_no
Police_station
location
Station_id
HAS
Officer_id
10
name
rank
Badge_no
Officer
registers
handles
Contact_n
age
Victim_ID
FIR_dat
Complaint_name
FIR_id
FIR
includes
Victim
Case_status
address
gender
Name
status
Prior_recordo
gender
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Crime_id
location
description
Crime_typ
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Criminal_id
name
Crime_data
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